



Gaming towards a cure

The University of Washington researchers create a puzzle game that helps coronavirus research. <https://digit.in/jun20-88>



Borderland Science

Borderlands 3 now has a arcade machines that let you play mini games that contribute to real research. <https://digit.in/jun20-89>

FEATURE

Berners-Lee, while he worked for CERN. The browser was actually called WorldWideWeb in the beginning, but later renamed to Nexus in order to not confuse it with the concept of the World Wide Web that was also an invention of Berners-Lee and others. While other concepts before it had sought to keep control over the data, and connections to servers by clients needed authentication and permission of server admins, the WWW and the browser worked together to do away with this.

Not only was the idea to openly share content adopted, but CERN also released it as a concept for anyone to adopt, with no royalties to be paid. Those who wanted to share information with the world were free to, and those who didn't could continue life unchanged. Everything we have now

as part of the net, can trace its origins back to the invention of the WWW and the browser, and once again it was someone, somewhere, deciding that profits were not all important.

MOORE'S LAW FOR EVERYTHING

Most people are aware of billionaire Gordon E. Moore, founder of Intel, and his prediction that the number of transistors on an IC would double every year, also called Moore's Law. However, most people aren't aware that his prediction was made in 1965, long before Intel was founded (1968). While Moore was talking specifically about transistors, his theory has become a meme of sorts and is used for pretty much everything in technology.

People have used Moore's Law to describe everything from the performance metrics of computers, the computational power of smartphones, the bandwidth of the average internet pipe in some countries, the amount of web pages on the internet, and many more things. If something in technology is getting better over time, by a predictable amount, a modified version of Moore's Law is tacked on, with date

ranges changing from the default of 'every two years' to every year, or every decade, and everything in between.

We are also using it in this same sense, and are using it to refer to technology complexity increasing at a steady pace over time – whether that's how much time we spend on our phones, how much work we can get done on a phone, etc. The achievement that we're celebrating isn't Moore's prediction, which was bordering on clairvoyant, if we're honest,



Moore's Law can be adapted to a heck of a lot of things!

but the collective drive from engineers all over the world, who work tirelessly to find new ways to improve every aspect of technology. They're so good at

their jobs, and have changed the world collectively, but don't get the recognition they deserve. It's the exploration of

new ideas, new manufacturing methods, miniaturisation of

everything that's enabled us to have on a phone, processing power that was considered to be a supercomputer under 30 years ago!

MOBILE COMPUTING

Speaking of phones, which are more powerful than laptops from a decade ago... it's not at all surprising that so many people don't even feel the need for another device to compute on. For better or worse, we now spend more time with our phones than anything else, and the younger the generation the more reliant they are on their phones for almost everything.

The concept of the touch screen, and multi-touch (not invented by Apple as many erroneously believe), is what changed the smartphone for good. Yes, the iPhone happened to be that device which did it the best, but there were many before it that tried and failed for one reason or the other. However,

once those tiny hardware keyboards were gotten rid of (much to BlackBerry's chagrin), it opened up the phone to essentially be one large screen. That's what made possible everything that we take for granted on a phone now.

Thanks to Moore's Law-ish development of processing power and miniaturisation, phones were finally powerful enough to accomplish much more than mere texting – even though that's a lot of what we do on a phone still. A convergence of a lot of technologies is what resulted in this complete shift in computing, and we will talk about a few of them next.

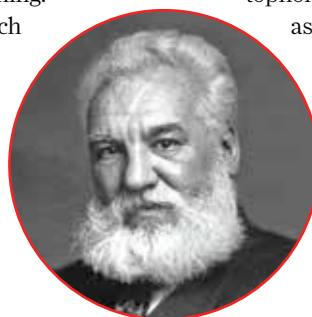
WIRELESS TECHNOLOGY

Whether we look at the wireless networking that we do at home or in offices, or the mobile networks that are pretty much all around us almost everywhere we go, one of the biggest shifts in the way we use technology was enabled by wireless.

Like all telephony, wireless also was born in the labs of Alexander Graham Bell, better known for inventing the telephone. Fans of a rival inventor called Elisha Gray will dispute this, citing the controversy around the patent of the telephone in 1875, but that's a rather long story for another time. Since Bell is still officially recognised as the first patent holder of the acoustic telegraph (later called the telephone), we're sticking to calling him that.

There is no controversy around who invented the photophone though, which was jointly worked on by Bell and his assistant Charles Tainter, and successfully worked in 1880. The photophone used the same principle

as a phone, but instead of sending a signal along a wire, it was sent on a beam of light. Even on his deathbed Bell proclaimed that the photophone, and not the telephone was his greatest ever invention, and we wholeheartedly agree. The photophone worked pretty much exactly the way we know fi-



Alexander Graham Bell's photophone was way ahead of it's time!